

Level 3 - Sorting Problems

1. Sort + Two Pointers

```
def find_pair(nums, target):  
  
    nums.sort()  
  
    left = 0  
    right = len(nums) - 1  
  
    while left < right:  
  
        total = nums[left] + nums[right]  
  
        if total == target:  
            return [nums[left], nums[right]]  
  
        elif total < target:  
            left += 1  
  
        else:  
            right -= 1  
  
    return []  
  
nums = [8, 3, 2, 7, 5]  
  
print(find_pair(nums, 10))
```

2. Sort + Duplicate Detection

```
def contains_duplicate(nums):  
  
    nums.sort()  
  
    for i in range(1, len(nums)):  
  
        if nums[i] == nums[i - 1]:  
            return True  
  
    return False  
  
nums = [5, 3, 8, 3, 1]  
  
print(contains_duplicate(nums))
```

3. Sort + Minimum Difference

```
def minimum_difference(nums):  
  
    nums.sort()  
  
    min_diff = float('inf')  
  
    for i in range(1, len(nums)):  
  
        diff = nums[i] - nums[i - 1]  
  
        min_diff = min(min_diff, diff)  
  
    return min_diff  
  
nums = [10, 3, 6, 20, 8]  
  
print(minimum_difference(nums))
```